A Wearable EEG Acquisition Device With Flexible Silver Ink Screen Printed Dry Sensors

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***Abstract*—Current electroencephalogram (EEG) measuring systems are bulky, impose constraints on patients, and require pre and post-measuring procedures. Usually, the EEG systems use either wet or dry EEG sensors, with the former suffers from skin preparation, the issues of adhesive conductive gels, and one-time usability whereas the latter causes skin irritation, abrasion, and pain upon pressure. Hence, these sensors are not suitable for long-term measurements. This paper presents a novel, wireless, behind-the-ear wearable EEG acquisition device that incorporates flexible dry EEG sensors. Silver ink-printed flexible sensors are fabricated using screen printing to overcome the above-mentioned drawbacks and limitations of conventional EEG sensors. The flexible sensors form a capacitive link with the skin via an adhesive layer between the sensor and the person's skin and are capable of acquiring the EEG without any skin preparation or gel. The performance of the printed flexible EEG sensors is tested by comparing them with the standard Ag/AgCl pre-gelled sensors. The alpha wave test and evoked potential EEG test are also performed for verification. The proposed device has a small form factor similar to a hearing aid and an in-house configurable Analog Front End (AFE) and Digital Back End (DBE) Processor and is capable of acquiring continuous EEG for a longer duration in a user-friendly and socially discrete manner.**

***Index Terms*—Analog Front End (AFE), Digital Back End (DBE), Chronic Neurological Disorders (CNDs), Flexible Dry EEG Sensors, Screen printing, Sensor-Skin Impedance (SSI), and Sensors (Electrodes).**

I. INTRODUCTION

An Electroencephalogram (EEG) is a non-invasive method of measuring brain activity by placing small metallic disk-shaped sensors on the scalp [1]. It plays a vital role in the diagnosis of chronic neurological disorders (CNDs) such as epilepsy, autism spectrum disorder (ASD) [3]- [5] and migraine [6]. It also helps in preventing further damage to the brain [7]. 24/7 continuous measurement of patient EEG for a longer duration is required for early, accurate diagnosis and treatment [7], [8].

Currently, there are medical wearable devices that are capable of monitoring brain activity but they are bulky, non-portable, ambulatory, or restricted to medical centers [9], [10], [12]. Contains long leads for connecting sensors with the recording system [11]. Have to sit or lay down during the recording procedures and in case of ambulatory setup have to carry heavy recording backpack [9]. Hence, patients are unable to perform daily routine activities. So there should be an unobtrusive wearable device for neural recordings while performing daily life activities [13].

Mostly, EEG systems use wet sensors for getting good quality brain signals that use a gel or saline but have certain disadvantages. This gel or electrolyte dries over a period of time and left residual which may result in allergy to some patients [14], [15]. Moreover, in the case of wet sensors, a trained technician is required for performing pre and post-measuring procedures including skin preparation, applying the adhesive conductive gel, placing sensors at standard EEG map positions, and in the end, cleansing [15], [16]. So it's more time-consuming and also results in a mess after usage [16]. Pre-gelled Ag/AgCl sensors are also used but they are one-time usable (up to a few hours). These wet EEG sensors can cause irritation, abrasions, and allergies onto the patient's skin upon long-term usage [17]. Similarly, the quality of the obtained signal gradually degrades with the drying and volatilization of electrolyte [14], [18].

Recently, some commercial EEG forehead bands are available that use dry metallic sheets like EEG sensors for the measurement and are developed to provide ease in wearing and maintain comfort level [19], [20]. Moreover, dry EEG sensors consisting of hard metallic pins structure are also used for EEG measurements in hairy sites by ensuring contact with the scalp. Both sensors are developed to overcome the issues related to wet sensors. However, these dry sensors have drawbacks i.e. upon application of pressure on the sensor, it will exert direct force onto the patient's skin resulting in pain and discomfort to the patients [15] - [17]. Moreover, the metallic sensors used in recent headbands usually form direct contact with the skin and may result in skin allergy, irritation, and abrasions [21]. Hence all these EEG sensors are not suitable for long-term measurements [22]. So, there is a dire need EEG system that mitigates the above-mentioned issues.

This work targets novel, flexible, silver ink-printed dry non-contact EEG sensors for acquiring EEG without any skin preparation or use of wet gel, along with a behind-the-ear wearable configurable EEG acquisition device having a small form factor [23]. The acquisition device is capable of removing motion artifacts and provides adjusting the overall gain to avoid amplifier saturation. The proposed device includes two major parts including Analog Front End (AFE) for EEG data acquisition with noise and motion artifacts removal and a digital back end (DBE) micro-controller for sending the real-

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time EEG data onto the mobile phone wirelessly.

II. EEG Sensors Design & Fabrication

Circular disk shape sensors are designed via screen printing of conductive silver (Ag) ink on a flexible Kapton polyimide film. Paron-910A conductive silver paste/conductive ink is used for the screen printing using the stencil. Fig. 1(a) shows the EEG sensors fabricated with the Ag Ink. Flexible EEG sensors with diameters of 4mm, 6mm, and 8mm are fabricated. There are three steps in the fabrication of the sensors a) printing of conductive Ag ink onto the Kapton polyimide substrate via screen printing using the stencil. b) Heat curing of the Ag ink that is applied to the fabricated sensors and finally c) the application of thin double-side adhesive tape on the (Ag Ink) printed side of the Kapton film to avoid direct contact of the metallic conductive layer with the skin. This layer acts as a dielectric and establishes a capacitive connection with the skin and provides a firm connection with the skin. As no conductive gel or hard bulky metallic sensor is present, moreover, these printed sensors are very thin and flexible resulting in making a good firm connection to the skin compared to conventional hard metallic sensors. Similarly, as a result of effective firm contact with the skin, there is a significant decrease in EEG noise. In Fig. 1 (b) flexibility of the developed EEG sensors is indicated. The cross-sectional view of the developed flexible EEG sensor is shown in Fig.1(c).

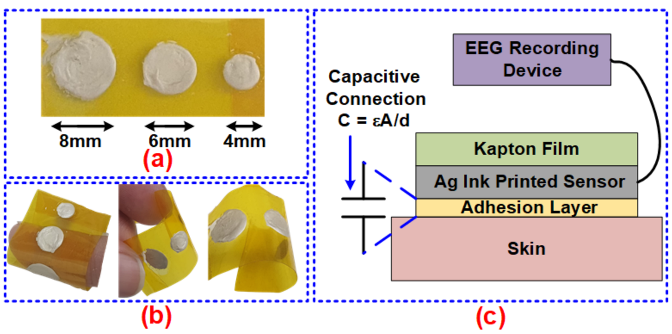


Fig. 1. (a) Various printed flexible EEG sensors on Kapton polyimide film. (b) The flexibility of the developed sensors. (c) Cross-sectional view of the printed EEG sensors.

Usually, a cable/lead is used to connect the sensors to the measuring device circuitry. Cables are capable of providing good quality bio-signals, but they are heavier in weight due to the connector clip and thick insulation layer [25]. Instead of using conventional cables, we have used a flexible printed connector (FPC) to electrically connect the EEG sensors with the acquisition circuit. Hence for designing the EEG sensors and connecting paths to be completely flexible, the acrylic sheet-based stencil is designed using laser cutting technology as shown in Fig. 2 (a). for printing the EEG band conveniently.

Fig.2 (b) shows the completely developed flexible EEG band consisting of the FPC attached to EEG sensors. FPC is fabricated along with the EEG sensors fabrication and follows the same procedure of screen printing onto the Kapton film using the stencil shown in Fig. 2 (a). One end of the FPC

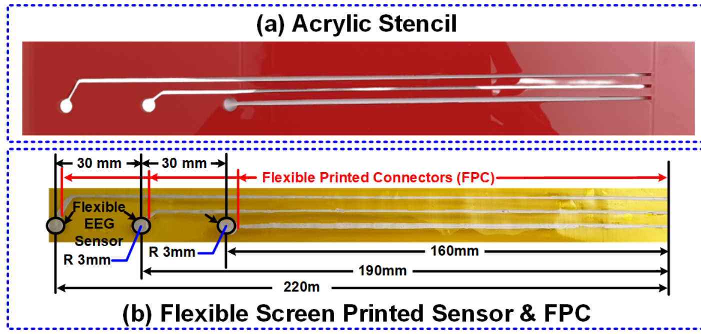


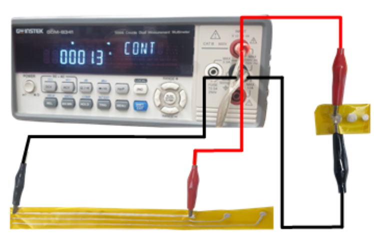
Fig. 2. (a) Acrylic sheet-based stencil for the screen printing of the complete EEG band incorporating the EEG sensor along with the FPC. (b) Complete flexible EEG band consisting of the flexible printed connector(FPC) along with the EEG sensors on Kapton polyimide film.

is connected to the EEG sensor and the other end will be connected to the acquisition device. These developed EEG sensors along with the FPC can easily be attached and removed from the acquisition device by plugging and unplugging from the connector pads inside the developed EEG acquisition device.

III. EEG Sensors Testing

A. Conductivity Test

The resistance of the printed flexible EEG sensors was measured using GWINSTEK Dual Measurement Multimeter, by attaching probes across the diameter of the printed side of the EEG sensor, before the application of the double-sided adhesive tape. Finally, the EEG sensors with a diameter of 6mm were used for the measurement. The measured resistance of various 6mm sensors is shown in Fig.3 and it shows the low resistance with a maximum value up to 1.3Ω. As FPC is used for connecting the sensors with the measuring device. So, the resistance of the whole setup including the EEG sensor and FPC is also measured, and the average value comes to 9.5Ω. Fig. 3 shows the resistance measuring setup along with the measured resistance. The results show the developed EEG sensors have sufficient conductivity along with low stable impedance.



Sensor	Resistance (Ω) EEG Sensor	Resistance (Ω) EEG Sensor + FPC
1	1.2	10.1
2	1.3	9.2
3	1.25	9.5
4	1.28	9.8
5	1.1	9.0

Fig. 3. Overall impedance measurement setup for the EEG sensor with and without flexible printed connector (FPC) & measured resistance values.

B. Skin-Sensor Contact Impedance Test (SSI)

Skin-sensor contact impedance Z_{SE-SK} plays an important role in the quality of obtained EEG signal [10]. Fig.4 (a)

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shows the complete equivalent impedance model of sensor-skin contact impedance. A large value of skin-sensor contact impedance Z_{SE-SK} degrades and attenuates the EEG signal [26]. In this test, a comparison between the skin-sensor contact impedance Z_{SE-SK} of our designed flexible dry EEG sensor and a conventional wet sensor is made.

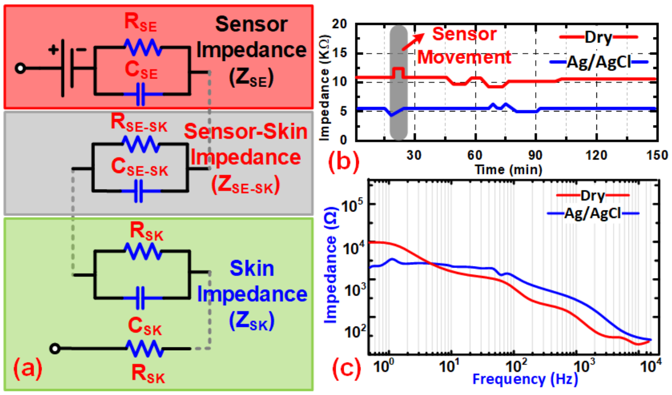

Fig. 4. (a) Complete skin-Sensor contact impedance (SSI) Equivalent Model. (b) Dry Flexible EEG sensor and Ag/AgCl pre-gelled sensors impedance measurement. (c) Variation of SSI with frequency.

The Neuroscan System was used to measure the skin-sensor contact impedance for a duration of 150 mins. Both our printed EEG sensors and standard pre-gelled Ag/AgCl sensors were placed close to each other. Fig. 4 (b) indicates the impedance measurements, An impedance of 5.6kΩ was obtained for Ag/AgCl pre-gelled sensors and an impedance of 10.3KΩ for our designed flexible sensor at FP2 location. Similarly, the effect of frequency variation on both sensors was observed as shown in Fig. Fig. 4(c). The skin-sensor contact impedance of the printed sensor is greater compared to Ag/AgCl pre-gelled electrodes but it is around 10kΩ which is sufficient for EEG acquisition [10], [27].

IV. EEG Acquisition Device

Having a miniaturized form factor along with user-friendly EEG sensors is critically important. So we have proposed an acquisition device having a form factor of a hearing aid. Hence remote monitoring and outside the clinical setting biosignals acquisition is feasible such as EEG while performing daily life routine activities and remote Brain-computer interface (BCI) is feasible using this device.

There are two major building blocks of the EEG acquisition device a) Analog Front End (AFE) and b) Digital Back End (DBE). AFE comprises of low noise amplifier (LNA) which consists of high input impedance instrumentation amplifier (IA). IA amplifies the differential EEG signal and offers a gain of 24dB, having a CMRR of 110 dB. After the IA, a band pass filter (BPF) having a low and high cut-off frequency of 0.16Hz and 48 Hz respectively, is added to get the desired informative EEG band and filter out the unnecessary noise and artifacts. Also, any leaked electrode DC offset is also being removed by this BPF. Then an operational amplifier is present which is a configurable programmable gain amplifier (PGA) and offers

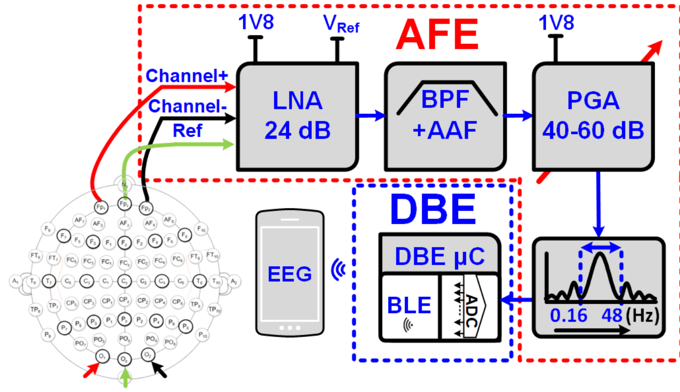

Fig. 5. System level diagram of the overall EEG acquisition system including AFE and DBE. 10 10 EEG tested positions are also indicated.

a variable gain of 40 dB to 60 dB to the EEG signal obtained from the IA. Hence, AFE acquires the EEG signal from the flexible EEG sensors, amplifies it, and removes noise and artifacts [23], [28]. A differential EEG channel is connected to the LNA input, significantly reducing the artifacts and power interference from the EEG Signal [24]. The electronics design of the EEG acquisition device is shown in Fig.5 consisting of AFE and DBE. A DBE is a u-blox NINAB316 ultra-low-power BLE integrated microcontroller. It converts the analog EEG signal into digital form using its 12-bit ADC block and sends the digitized EEG data to the mobile phone wirelessly.

Behind-the-ear wearable with a minimal form factor similar to a hearing aid EEG acquisition device is designed. It's an ultra-low power, socially discrete user-friendly system, powered up via a rechargeable battery (3.7V Li-Ion 300mAh) and fully operational during the charging phase. A standard micro type-B USB connector is provided for charging. A PCB prototype is developed using commercial off-the-shelf components. The casing enclosure is designed via 3D resin printing. Fig.6 shows the completely assembled device along with its details. The proposed wearable device using the printed EEG sensors applied on the person for testing and measurements indicated in Fig.6.

Fig. 7 depicts the major building blocks of the proposed EEG device and sensor. The Kapton film-based EEG sensor band consists of differential channel sensors that can be placed on the Fp1, Fp2, O1, and O2 positions of the 10 10 EEG system, and reference channel placed on the Fpz or Oz position. EEG acquisition consists of AFE, DBE, and Power management indicated in Fig. 7. The device's 3D enclosure contains two parts, a top and bottom casing with the ear clip along with the bottom part.

V. EEG Measurement

A. Simultaneous EEG Measurement

EEG Signal is measured using both the proposed dry flexible sensors and Ag/AgCl pre-gelled sensors simultaneously from the brain's frontal lobe. It is observed that dry flexible sensors show promising results compared to standard sensors. Fig.8. shows the EEG signals obtain using the proposed

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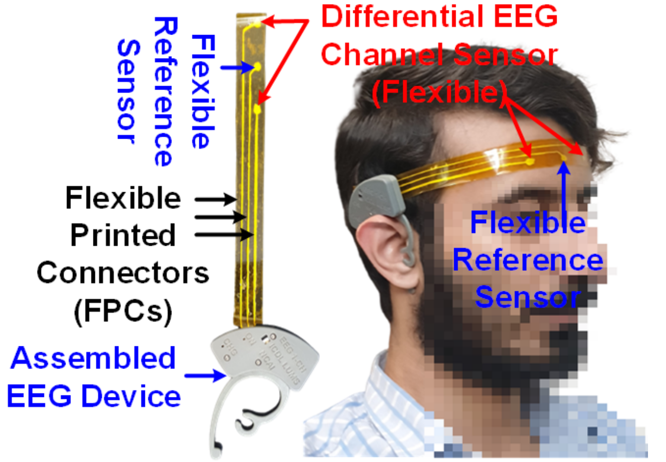

Fig. 6. The complete assembled EEG device along with its application on the person for EEG measurement.

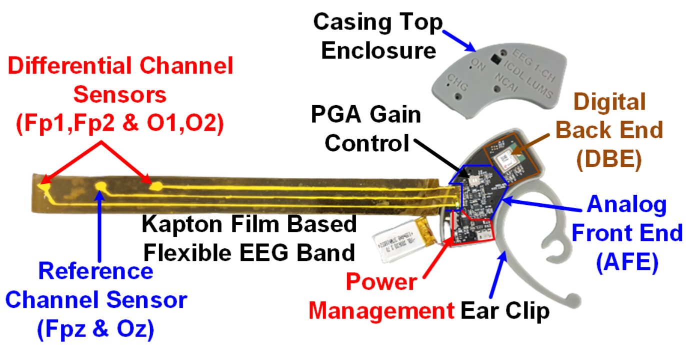

Fig. 7. Building blocks of the proposed EEG acquisition device along with the flexible EEG sensor band.

printed EEG Sensors (blue) and standard EEG sensors (red) with the eye blink activities (both eyes blinks (BEB), right eye blink (REB)) indicated by the grey-shaded regions from the frontal lobe.

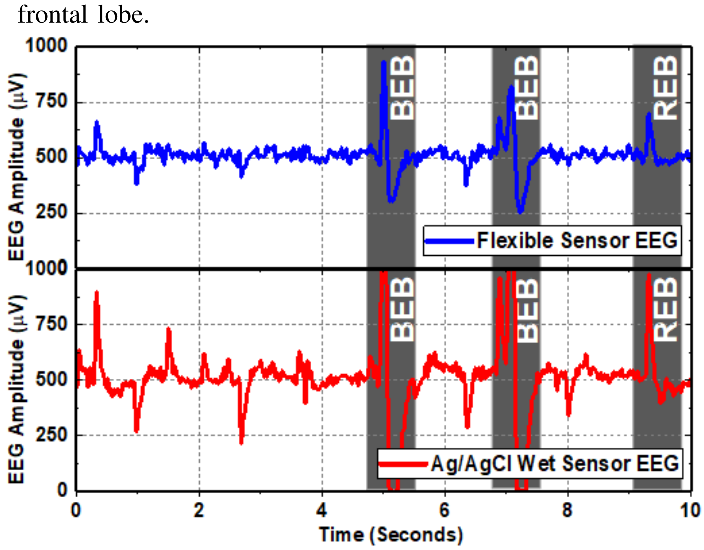

Fig. 8. EEG measurement using flexible dry fabricated sensors (blue) and Ag/AgCl wet sensors (red) at the frontal lobe.

B. Visual Evoked Potential Test

The designed EEG sensors are also tested on the occipital lobe and Visual Evoked Potential (VEP) test is conducted with a stimulus frequency of 13 Hz. Fig. 9 shows a 30s duration VEP test EEG measured from the occipital lobe having a stimulus frequency of 13Hz from a flickering screen in a noise and distraction-free room.

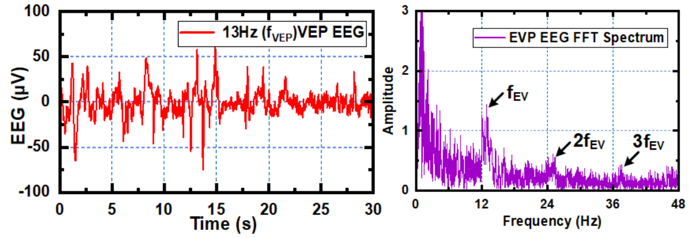

Fig. 9. 30s EEG signal under 13Hz VEP stimulus is being applied from the computer screen. FFT of the EEG signal clearly indicates the peak at a frequency of 13Hz and frequency multiple of the fundamental frequency f_{VEP}.

C. Alpha Wave Test

In order to verify the proposed flexible EEG sensors along with the EEG acquisition device, the alpha wave test is also applied using the occipital lobe, and results are shown in Fig. 10.

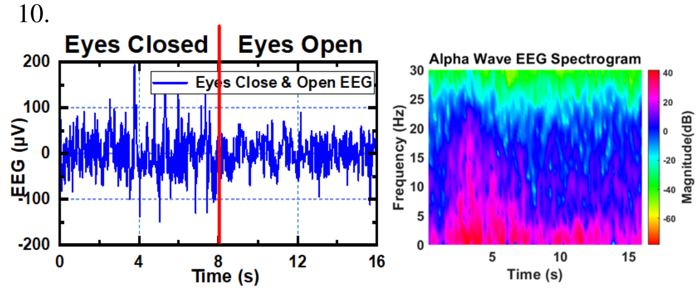

Fig. 10. Alpha wave test, Eyes closed for first 8s and open for the last 8s. The respective spectrogram is also indicated for the evaluation of alpha rhythm during the closed-eyes state.

VI. Conclusion

A wearable EEG acquisition device with novel flexible, Ag ink screen printed dry sensors is presented. No skin preparation or gel is required for the signal acquisition from the EEG sensors. The designed sensors are flexible, non-contact, non-irritating to the skin, comfortable to wear and cause no pain or any inconvenience to the user. Similarly, the EEG acquisition device is of small form factor, behind-the-ear wearable. The performance of the proposed dry flexible EEG sensors is compared with standard Ag/AgCl pre-gelled sensors and provides good results. The EEG acquisition with the proposed dry flexible sensors is a major step forward in a socially discrete manner and initiative for outside the clinical setting EEG acquisition.

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